

This exercise sheet provides some training material for the optional tutorial. Some of the exercises and solutions have been taken from Wohlgemuth's book.

Exercise 1

Translate each of the following statements into a first-order formula using $\in$ as the only predicate symbol:

- a) If $A \subseteq B$, then A and $C \setminus B$ are disjoint.
- b) A is not a subset of B .
- c) $\neg(A \cup B \subseteq C \setminus D)$.

Solution 1

- a) If $A \subseteq B$, then A and $C \setminus B$ are disjoint.

The desired formula is an implication. Its left hand side is $\forall x (x \in A \rightarrow x \in B)$. Its right hand side is as follows.

If the two sets A and $C \setminus B$ are disjoint, then they have no element in common, i.e., there is no element which occurs in both sets. The formula $\neg \exists x (x \in A \wedge x \in C \setminus B)$ represents this fact. Since $C \setminus B = \{x \mid x \in C \wedge x \notin B\}$, we get $\neg \exists x (x \in A \wedge x \in C \wedge x \notin B)$ for the right hand side. The overall solution is therefore

$$(\forall x (x \in A \rightarrow x \in B)) \rightarrow (\neg \exists x (x \in A \wedge x \in C \wedge x \notin B)).$$

- b) A is not a subset of B .

Then we have $\neg(A \subseteq B)$ which is equivalent to $\neg \forall x (x \in A \rightarrow x \in B)$. This is in turn equivalent to $\exists x (x \in A \wedge x \notin B)$.

- c) $\neg(A \cup B \subseteq C \setminus D)$.

Start with $\neg \forall x ((x \in A \cup B) \rightarrow (x \in C \setminus D))$ and derive the following solution:

$$\exists x ((x \in A \vee x \in B) \wedge (x \notin C \vee x \in D))$$

Exercise 2

Eliminate def^2 from the proof fragment below.

- 1. $a \in M$
- 2. $M \subseteq N$
- 3. $a \in N$ (1,2; $\text{def}^2 \subseteq$)

Solution 2

1. $a \in M$
2. $M \subseteq N$
3. for all $t \in M: t \in N$ (2; def $\subseteq$)
3. $a \in N$ (1,3; def $\subseteq$)

Exercise 3

Let A and B be sets. Define $A \setminus B = \{x \mid x \in A \wedge x \notin B\}$ and $\overline{B} = \{x \in U \mid x \notin B\}$, where U is the universal set. Prove $(A \setminus B) = (A \cap \overline{B})$ in the following two steps.

- a) Show: $(A \setminus B) \subseteq (A \cap \overline{B})$.
- b) Show: $(A \cap \overline{B}) \subseteq (A \setminus B)$.

Solution 3

a)

1. Let $x \in A \setminus B$ be arbitrary
2. $x \in A \wedge x \notin B$ (1; def $\setminus$)
3. $x \in A$ (2; us $\wedge$)
4. $x \notin B$ (2; us $\wedge$)
5. $x \in \overline{B}$ (4; def $\overline{B}$)
6. $x \in A \cap \overline{B}$ (3,5; def $\cap$)
7. for all x s.t. $x \in A \setminus B: x \in A \cap \overline{B}$ (1 – 6; pr $\forall$)
8. $A \setminus B \subseteq A \cap \overline{B}$ (7; def $\subseteq$)

b)

1. Let $x \in A \cap \overline{B}$ be arbitrary
2. $x \in A$ (1; us $\cap$)
3. $x \in \overline{B}$ (2; us $\cap$)
4. $x \notin B$ (3; def $\overline{B}$)
5. $x \in A \wedge x \notin B$ (2,4; pr $\wedge$)
6. $x \in A \setminus B$ (5; def $\setminus$)
7. for all x s.t. $x \in A \cap \overline{B}: x \in A \setminus B$ (1 – 6; pr $\forall$)
8. $A \cap \overline{B} \subseteq A \setminus B$ (7; def $\subseteq$)

Exercise 4

Show: $\forall$ sets $A \forall$ sets $B: ((A \setminus B) \cup B) = (A \cup B)$. Contrary to the examples in the lecture, handle the forall statements on the object level and not on the meta level.

Solution 4

We decompose the proof into two subproofs, namely

1. the proof of $\forall$ sets $A \forall$ sets $B: ((A \setminus B) \cup B) \subseteq (A \cup B)$ and
2. the proof of $\forall$ sets $A \forall$ sets $B: (A \cup B) \subseteq ((A \setminus B) \cup B)$.

The above statement follows from 1. $\wedge$ 2. by quantifier rules and the definition of $=$.

1.	Let A be an arbitrary set	
2.	Let B be an arbitrary set	
3.	Let $x \in (A \setminus B) \cup B$ be arbitrary	
4.	$x \in A \setminus B$ or $x \in B$	(3; def $\cup$)
5.	Case 1: $x \in A \setminus B$	
6.	$x \in A$	(5; def $\setminus$)
7.	$x \in A \cup B$	(6; def $\cup$)
8.	Case 2: $x \in B$	
9.	$x \in A \cup B$	(8; def $\cup$)
10.	$x \in A \cup B$	(4 – 9; us $\vee$)
11.	forall x s.t. $x \in (A \setminus B) \cup B: x \in A \cup B$	(3 – 10; pr $\forall$)
12.	$(A \setminus B) \cup B \subseteq A \cup B$	(11; def $\subseteq$)
13.	forall sets $B: (A \setminus B) \cup B \subseteq A \cup B$	(2 – 12; pr $\forall$)
14.	forall sets $A: (\text{forall sets } B: (A \setminus B) \cup B \subseteq A \cup B)$	(1 – 13; pr $\forall$)

1.	Let A be an arbitrary set	
2.	Let B be an arbitrary set	
3.	Let $x \in A \cup B$ be arbitrary	
4.	$x \in A$ or $x \in B$	(3; def $\cup$)
5.	Case 1: $x \in A$	
6.	$x \in B$ or $x \notin B$	(lem)
7.	Case 1.1: $x \in B$	
8.	$x \in (A \setminus B) \cup B$	(7; def $\cup$)
9.	Case 1.2: $x \notin B$	
10.	$x \in (A \setminus B)$	(5, 9; def $\setminus$)
11.	$x \in (A \setminus B) \cup B$	(10; def $\cup$)
12.	$x \in (A \setminus B) \cup B$	(6 – 11; us $\vee$)
13.	Case 2: $x \in B$	
14.	$x \in (A \setminus B) \cup B$	(13; def $\cup$)
15.	$x \in (A \setminus B) \cup B$	(4 – 14; us $\vee$)
16.	$A \cup B \subseteq (A \setminus B) \cup B$	(3 – 15; def $\subseteq$)
17.	forall sets $B: (A \cup B \subseteq (A \setminus B) \cup B)$	(2 – 16; pr $\forall$)
18.	forall sets $A: (\text{forall sets } B: (A \cup B \subseteq (A \setminus B) \cup B))$	(1 – 17; pr $\forall$)

Exercise 5

Show: $\overline{\bigcup_{i=1}^n A_i} = \bigcap_{i=1}^n \overline{A_i}$ holds for sets $A_1, \dots, A_n$.

Solution 5

1.	Let $x \in \overline{\bigcup_{i=1}^n A_i}$ be arbitrary	
2.	$x \notin \bigcup_{i=1}^n A_i$	(1; def $(\overline{\cdot})$)
3.	$\bigwedge_{i=1}^n x \notin A_i$	(2; def $\cup$)
4.	$\bigwedge_{i=1}^n x \in \overline{A_i}$	(3; def $(\overline{\cdot})$)
5.	$x \in \bigcap_{i=1}^n \overline{A_i}$	(4; def $\cap$)
6.	$\overline{\bigcup_{i=1}^n A_i} \subseteq \bigcap_{i=1}^n \overline{A_i}$	(1 – 5; def $\subseteq$)
7.	Let $x \in \bigcap_{i=1}^n \overline{A_i}$ be arbitrary	
8.	$\bigwedge_{i=1}^n x \in \overline{A_i}$	(7; def $\cap$)
9.	$\bigwedge_{i=1}^n x \notin A_i$	(8; def $(\overline{\cdot})$)
10.	$x \notin \bigcup_{i=1}^n A_i$	(9; def $\cup$)
11.	$x \in \overline{\bigcup_{i=1}^n A_i}$	(10; def $(\overline{\cdot})$)
12.	$\bigcap_{i=1}^n \overline{A_i} \subseteq \overline{\bigcup_{i=1}^n A_i}$	(7 – 11; def $\subseteq$)
13.	$\overline{\bigcup_{i=1}^n A_i} = \bigcap_{i=1}^n \overline{A_i}$	(6, 12; def $=$)

Exercise 6

Assume that A , B , and C are sets. Show: $A \cap (B \cup C) = (A \cap B) \cup (A \cap C)$.

Solution 6

1.	Let $x \in A \cap (B \cup C)$ be arbitrary	
2.	$x \in A$	(1; def $\cap$)
3.	$x \in B \cup C$	(1; def $\cap$)
4.	$x \in B$ or $x \in C$	(3; def $\cup$)
5.	Case 1: $x \in B$	
6.	$x \in A \cap B$	(2, 5; def $\cap$)
7.	$x \in A \cap B \cup A \cap C$	(6; def $\cup$)
8.	Case 2: $x \in C$	
9.	$x \in A \cap C$	(2, 8; def $\cap$)
10.	$x \in A \cap B \cup A \cap C$	(9; def $\cup$)
11.	$x \in A \cap B \cup A \cap C$	(4 – 10; pr $\vee$)
12.	$A \cap (B \cup C) \subseteq A \cap B \cup A \cap C$	(1 – 11; def $\subseteq$)
13.	Let $x \in A \cap B \cup A \cap C$ be arbitrary	
14.	$x \in A \cap B$ or $x \in A \cap C$	(13; def $\cup$)
15.	Case 1: $x \in A \cap B$	
16.	$x \in A$	(15; def $\cap$)
17.	$x \in B$	(15; def $\cap$)
18.	$x \in B \cup C$	(17; def $\cup$)
19.	$x \in A \cap (B \cup C)$	(16, 18; def $\cap$)
20.	Case 2: $x \in A \cap C$	
21.	$x \in A$	(20; def $\cap$)
22.	$x \in C$	(20; def $\cap$)
23.	$x \in B \cup C$	(22; def $\cup$)
24.	$x \in A \cap (B \cup C)$	(21, 23; def $\cap$)
25.	$x \in A \cap (B \cup C)$	(14 – 24; pr $\vee$)
26.	$A \cap B \cup A \cap C \subseteq A \cap (B \cup C)$	(13 – 25; def $\subseteq$)
27.	$A \cap (B \cup C) = A \cap B \cup A \cap C$	(12, 26; def $=$)

Exercise 7

For sets A , B , and C :

- a) Show: $A \subseteq B$ iff $A \cap B = A$.
b) Show: $A \subseteq B$ iff $A \cup B = B$.

Solution 7

a) We show the iff in two parts, namely $A \subseteq B \rightarrow A \cap B = A$ and $A \cap B = A \rightarrow A \subseteq B$.

1.	$A \subseteq B$	
2.	Let $x \in A \cap B$ be arbitrary	
3.	$x \in A$	(2; def $\cap$)
4.	$A \cap B \subseteq A$	(2, 3; def $\subseteq$)
5.	Let $x \in A$ be arbitrary	
6.	$x \in B$	(1, 5; def $\subseteq$)
7.	$x \in A \cap B$	(5, 6; def $\cap$)
8.	$A \subseteq A \cap B$	(5 – 7; def $\subseteq$)
9.	$A \cap B = A$	(4, 8; def $=$)
10.	if $A \subseteq B$ then $A \cap B = A$	(1 – 9; pr $\rightarrow$)

1.	$A \cap B = A$	
2.	Let $x \in A$ be arbitrary	
3.	$x \in A \cap B$	(1, 2; substitution)
4.	$x \in B$	(3; def $\cap$)
5.	$A \subseteq B$	(2 – 4; def $\subseteq$)
6.	if $A \cap B = A$ then $A \subseteq B$	(1 – 5; pr $\rightarrow$)

b) We show the iff in two parts, namely $A \subseteq B \rightarrow A \cup B = B$ and $A \cup B = B \rightarrow A \subseteq B$.

1.	$A \subseteq B$	
2.	Let $x \in A \cup B$ be arbitrary	
3.	$x \in A$ or $x \in B$	(2; def $\cup$)
4.	Case 1: $x \in A$	
5.	$x \in B$	(1, 4; def $\subseteq$)
6.	Case 2: $x \in B$	
7.	$x \in B$	(3 – 6; pr $\vee$)
8.	$A \cup B \subseteq B$	(2 – 7; def $\subseteq$)
9.	Let $x \in B$ be arbitrary	
10.	$x \in A \cup B$	(9; def $\cup$)
11.	$B \subseteq A \cup B$	(9 – 10; def $\subseteq$)
12.	$A \cup B = B$	(8, 11; def $=$)
13.	if $A \subseteq B$ then $A \cup B = B$	(1 – 12; pr $\rightarrow$)

1.	$A \cup B = B$	
2.	Let $x \in A$ be arbitrary	
3.	$x \in A \cup B$	(2; def $\cup$)
4.	$x \in B$	(1, 3; subst)
5.	$A \subseteq B$	(2 – 4; def $\subseteq$)
6.	if $A \cup B = B$ then $A \subseteq B$	(1 – 5; pr $\rightarrow$)

Exercise 8

Given sets A_k (for $k = 1, 2, 3, 4$), define

$$\begin{aligned} B_1 &= A_1 & B_2 &= A_2 \setminus A_1 \\ B_3 &= A_3 \setminus (A_1 \cup A_2) & B_4 &= A_4 \setminus (A_1 \cup A_2 \cup A_3). \end{aligned}$$

- a) Prove: $\bigcup_{i=1}^4 B_i = \bigcup_{i=1}^4 A_i$
- b) Prove: For all $1 \leq i, j \leq 4, i \neq j$: $B_i \cap B_j = \emptyset$

Solution 8

Recall from above that (*) $A \setminus B = A \cap \overline{B}$ holds.

- a) We first show the following lemma: For any sets A, B , $(A \setminus B) \cup B = A \cup B$ holds. Contrary to above, we use (*), we omit the quantifier rules and we put everything in one proof.

Proof of the lemma:

1.	Let $x \in (A \setminus B) \cup B$ be arbitrary	
2.	$x \in (A \cap \overline{B}) \cup B$	(1, (*); subst)
3.	$x \in (A \cap \overline{B})$ or $x \in B$	(2; def $\cup$)
4.	Case 1: $x \in A \cap \overline{B}$	
5.	$x \in A$	(4; def $\cap$)
6.	$x \in A \cup B$	(5; def $\cup$)
7.	Case 2: $x \in B$	
8.	$x \in A \cup B$	(7; def $\cup$)
9.	$x \in A \cup B$	(3 – 8; pr $\vee$)
10.	$(A \setminus B) \cup B \subseteq A \cup B$	(1 – 9; def $\subseteq$)
11.	Let $x \in A \cup B$ be arbitrary	
12.	$x \in A$ or $x \in B$	(11; def $\cup$)
13.	Case 1: $x \in A$	
14.	$x \in B$ or $x \notin B$	(lem)
15.	Case 1.1: $x \in B$	
16.	$x \in (A \setminus B) \cup B$	(15; def $\cup$)
17.	Case 1.2: $x \notin B$	
18.	$x \in (A \setminus B)$	(13, 17; def $\setminus$)
19.	$x \in (A \setminus B) \cup B$	(18; def $\cup$)
20.	$x \in (A \setminus B) \cup B$	(14 – 19; pr $\vee$)
21.	Case 2: $x \in B$	
22.	$x \in (A \setminus B) \cup B$	(21; def $\cup$)
23.	$x \in (A \setminus B) \cup B$	(12 – 22; pr $\vee$)
24.	$A \cup B \subseteq (A \setminus B) \cup B$	(11 – 23; def $\subseteq$)
25.	$(A \setminus B) \cup B = A \cup B$	(10, 24; def $=$)

Next we show $\bigcup_{i=1}^4 B_i = \bigcup_{i=1}^4 A_i$ We use A_k^* to denote $\bigcup_{i=1}^{k-1} A_i$ for $k = 2, 3, 4$.

1.	Let $x \in B_1 \cup B_2 \cup B_3 \cup B_4$ be arbitrary	
2.	$x \in A_1 \cup \bigcup_{i=2}^4 (A_i \setminus A_i^*)$	(1; def B_i)
3.	$x \in A_1 \cup \bigcup_{i=2}^4 (A_i \cup A_i^*)$	(2; apply lemma)
4.	$x \in A_1 \cup A_2 \cup A_3 \cup A_4$	(3; prop $\cup$)
5.	$\bigcup_{i=1}^4 B_i \subseteq \bigcup_{i=1}^4 A_i$	(1 – 4; def $\subseteq$)
6.	Let $x \in A_1 \cup A_2 \cup A_3 \cup A_4$ be arbitrary	
7.	$x \in A_1 \cup \bigcup_{i=2}^4 (A_i \cup A_i^*)$	(6; prop $\cup$)
8.	$x \in A_1 \cup \bigcup_{i=2}^4 (A_i \setminus A_i^*)$	(7; apply lemma)
9.	$x \in B_1 \cup B_2 \cup B_3 \cup B_4$	(8; def B_i)
10.	$\bigcup_{i=1}^4 A_i \subseteq \bigcup_{i=1}^4 B_i$	(6 – 9; def $\subseteq$)
11.	$\bigcup_{i=1}^4 B_i = \bigcup_{i=1}^4 A_i$	(5, 10; def $=$)

- b) Recall from above that $(**) \overline{\bigcup_{i=1}^n A_i} = \bigcap_{i=1}^n \overline{A_i}$ holds for sets $A_1, \dots, A_n$. Let $F(i, j)$ denote $B_i \cap B_j = \emptyset$ and observe that $F(i, j)$ is symmetric to $F(j, i)$ due to the commutativity of $\cap$. Our statement to prove can be reformulated as

$$\forall i \forall j ((1 \leq i < j \leq 4) \implies F(i, j)) \wedge \forall i \forall j ((1 \leq i < j \leq 4) \implies F(j, i)).$$

The two conjunctively connected proof tasks are symmetric and it is sufficient to prove the first one. We use $A_1^* = \emptyset$ and A_k^* to denote $\bigcup_{i=1}^{k-1} A_i$ for $k = 2, 3, 4$. Then B_i ($1 \leq i \leq 4$) can be rewritten as $A_i \setminus A_i^*$. Recall that $A_1 \setminus \emptyset$ is A_1 , $\overline{\emptyset}$ is the universal set U and $A \cap U = A$ for any set A .

1.	Let i, j with $1 \leq i < j \leq 4$ be arbitrary	
2.	Let $x \in B_i \cap B_j$ be arbitrary	
3.	$x \in (A_i \setminus A_i^*) \cap (A_j \setminus A_j^*)$	(2, subst def B_i, B_j)
4.	$x \in (A_i \cap \overline{A_i^*}) \cap (A_j \cap \overline{A_j^*})$	(3, $(*)$ subst)
5.	$x \in (A_i \cap \bigcap_{k=1}^{i-1} \overline{A_k}) \cap (A_j \cap \bigcap_{l=1}^{j-1} \overline{A_l})$	(4, lemma $(**)$)
6.	$x \in \emptyset$	(5, because $A_i \cap \overline{A_i} = \emptyset$)
7.	$B_i \cap B_j \subseteq \emptyset$	(2 – 6; def $\subseteq$)
8.	$\emptyset \subseteq B_i \cap B_j$	(property of sets)
9.	$B_i \cap B_j = \emptyset$	(7, 8; def $=$)
10.	forall i, j with $1 \leq i < j \leq 4$: $B_i \cap B_j = \emptyset$	(1 – 9; pr $\forall$)

Exercise 9

Let $f: A \mapsto B$. Suppose $X \subseteq A$, $Y \subseteq B$, and $y \in B$. Define:

$$\text{P1 } f(X) = \{b \in B \mid b = f(x) \text{ for some } x \in X\}$$

$$\text{P2 } f^{-1}(Y) = \{a \in A \mid f(a) \in Y\}$$

$$\text{P3 } f^{-1}(y) = \{a \in A \mid f(a) = y\}$$

Let $C \subseteq B$, and $D \subseteq B$. Show: $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$. First structure the proof in the same way as we discussed in the lecture. Then present the proof in English.

Solution 9

Assume:	1.	$C \subseteq B$	
	2.	$D \subseteq B$	
Show:		$f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$	
	1.	Let $x \in f^{-1}(C \cap D)$	
	2.	$f(x) \in C \cap D$	(1; def $f^{-1}(\cdot)$, P2)
	3.	$f(x) \in C$	(2; def $\cap$)
	4.	$f(x) \in D$	(2; def $\cap$)
	5.	$x \in f^{-1}(C)$	(3; def $f^{-1}(\cdot)$, P2)
	6.	$x \in f^{-1}(D)$	(4; def $f^{-1}(\cdot)$, P2)
	7.	$x \in f^{-1}(C) \cap f^{-1}(D)$	(5, 6; def $\cap$)
	8.	$f^{-1}(C \cap D) \subseteq f^{-1}(C) \cap f^{-1}(D)$	(1–7; def $\subseteq$)
	9.	Let $x \in f^{-1}(C) \cap f^{-1}(D)$	
	10.	$x \in f^{-1}(C)$	(9; def $\cap$)
	11.	$x \in f^{-1}(D)$	(9; def $\cap$)
	12.	$f(x) \in C$	(10; def $f^{-1}(\cdot)$, P2)
	13.	$f(x) \in D$	(11; def $f^{-1}(\cdot)$, P2)
	14.	$f(x) \in C \cap D$	(12, 13; def $\cap$)
	15.	$x \in f^{-1}(C \cap D)$	(14; def $f^{-1}(\cdot)$, P2)
	16.	$f^{-1}(C) \cap f^{-1}(D) \subseteq f^{-1}(C \cap D)$	(9–15; def $\subseteq$)
	17.	$f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$	(8, 16; def $=$)

An English version is as follows (cf Wohlgemuth's book pp. 115–116).

Assume $C \subseteq B$ and $D \subseteq B$. We will show that $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$ holds.

First we show that

$$(i) \quad f^{-1}(C \cap D) \subseteq f^{-1}(C) \cap f^{-1}(D)$$

holds. Let x be an element of $f^{-1}(C \cap D)$. Then $f(x) \in C \cap D$ by P2 and therefore $f(x) \in C$ as well as $f(x) \in D$ by the definition of set intersection. Hence, by P2, $x \in f^{-1}(C)$ as well as $x \in f^{-1}(D)$ holds. It follows by the definition of set intersection that $x \in f^{-1}(C) \cap f^{-1}(D)$ holds. Consequently $f^{-1}(C \cap D) \subseteq f^{-1}(C) \cap f^{-1}(D)$ holds by the definition of the subset relation.

Second we show that

$$(ii) \quad f^{-1}(C) \cap f^{-1}(D) \subseteq f^{-1}(C \cap D)$$

holds. Let x be an element of $f^{-1}(C) \cap f^{-1}(D)$. By the definition of set intersection, x is an element of $f^{-1}(C)$ and x is an element of $f^{-1}(D)$. By P2, $f(x)$ is an element of C and of D . But then $f(x)$ is an element of $C \cap D$ by the definition of set intersection. Again by P2, x is also in $f^{-1}(C \cap D)$. Consequently $f^{-1}(C) \cap f^{-1}(D) \subseteq f^{-1}(C \cap D)$ holds by the definition of the subset relation.

From (i) and (ii) together with the definition of equality, we obtain $f^{-1}(C \cap D) = f^{-1}(C) \cap f^{-1}(D)$ as desired.

Exercise 10

Prove or refute the following:

Let $f: A \mapsto B$, $E \subseteq A$, $F \subseteq A$. Then $f(E \cap F) = f(E) \cap f(F)$.

Solution 10

The statement does *not* hold. Consider the following counter example. Let $f: \{1, 2, 3\} \mapsto \{4, 5\}$ be defined by $f(1) = 4$, $f(2) = 5$, $f(3) = 4$. Let $E = \{1, 2\}$ and $F = \{2, 3\}$. Then $E \cap F = \{2\}$ and $f(E \cap F) = \{5\}$. Consider $f(E) = \{4, 5\}$ and $f(F) = \{4, 5\}$. Then $f(E) \cap f(F) = \{4, 5\}$ which is different from $f(E \cap F)$.

Exercise 11

A function $f: A \mapsto B$ is called *one-to-one* provided that for all $a_1, a_2 \in A$: if $f(a_1) = f(a_2)$ then $a_1 = a_2$.

Let $f: \mathbb{R} \mapsto \mathbb{R}$ be defined by $f(x) = 2x + 4$ for all $x \in \mathbb{R}$. Prove that f is one-to-one.

Solution 11

- | | | |
|----|---|----------------------------------|
| 1. | Let $x, y \in \mathbb{R}$ and $f(x) = f(y)$ | |
| 2. | $2x + 4 = 2y + 4$ | (1; def of f) |
| 3. | $2x = 2y$ | (2; prop $\mathbb{R}$) |
| 4. | $x = y$ | (3; prop $\mathbb{R}$) |
| 5. | for all $x, y \in \mathbb{R}$: if $f(x) = f(y)$, then $x = y$ | (1–4; pr $\forall \rightarrow$) |
| 6. | f is one-to-one | (5; def one-to-one) |

Exercise 12

A function $f: A \mapsto B$ is called *onto* if for each $b \in B$ there exists some $a \in A$ such that $f(a) = b$.

Show that there exists a one-to-one function $g: \mathbb{N} \mapsto \mathbb{N}$ which is not onto.

Solution 12

Creativity is needed because the function g has to be invented. Define $g: \mathbb{N} \mapsto \mathbb{N}$ by $g(n) = n + 1$ for all $n \in \mathbb{N}$. Similar to above, one can show that g is one-to-one (proof omitted).

For the sake of contradiction, assume that g is onto. Then $g(m) = 1$ for some $m \in \mathbb{N}$. Thus $m + 1 = 1$ and $m = 0$. The latter statement contradicts that $m \in \mathbb{N}$.

Remark: In the proof, it is assumed that $0 \notin \mathbb{N}$. What do we have to change when $\mathbb{N}$ contains 0?

Exercise 13

A *sequence* is a function $a: \mathbb{N} \mapsto \mathbb{R}$. $a(n)$ is called the n th term of the sequence and is denoted by a_n . The sequence itself is denoted by $\langle a_1, a_2, \dots \rangle$ or by $\langle a_n \rangle$. A number $L \in \mathbb{R}$ is defined to be the *limit* of $\langle a_n \rangle$, written $\lim_{n \rightarrow \infty} \langle a_n \rangle = L$, provided that, given a real number $\epsilon > 0$, there exists a natural number N such that for all $n > N$: $|a_n - L| < \epsilon$. If L is the limit of $\langle a_n \rangle$, then $\langle a_n \rangle$ *converges* to L . A sequence with no limit is said to *diverge*.

Show that $\langle a_n \rangle$ defined by $a_n = n/(n+1)$ has limit 1.

Solution 13

- | | | |
|----|---|----------------------------|
| 1. | Let $\epsilon > 0$ be given | |
| 2. | $N > \frac{1}{\epsilon}$ for some $N \in \mathbb{N}$ | (Arch. prop $\mathbb{R}$) |
| 3. | Let n be arbitrary such that $n > N$ | |
| 4. | $\frac{1}{\epsilon} < N < n < (n+1)$ | (2, 3; prop $\mathbb{R}$) |
| 5. | $1 < \epsilon(n+1)$ | (4; prop $\mathbb{R}$) |
| 6. | $ n - (n+1) < \epsilon(n+1)$ | (5; prop $\mathbb{R}$) |
| 7. | $ \frac{n}{n+1} - 1 < \epsilon$ | (6; prop $\mathbb{R}$) |
| 8. | for all $n > N$: $ \frac{n}{n+1} - 1 < \epsilon$ | (3–7, pr $\forall$) |
| 9. | $\lim_{n \rightarrow \infty} \langle a_n \rangle = 1$ | (1–8, def limit) |

We used the *Archimedean property* of $\mathbb{R}$, i.e., for any $r \in \mathbb{R}$, there exists $n \in \mathbb{N}$ such that $r < n$.

Exercise 14

Let $\langle a_n \rangle$ be a sequence. If $\lim_{n \rightarrow \infty} \langle a_n \rangle = L$ and $\lim_{n \rightarrow \infty} \langle a_n \rangle = M$, then $L = M$. That is, if a sequence has a limit, then it is unique.

Hint: $|x - z| \leq |x - y| + |y - z|$ for all $x, y, z \in \mathbb{R}$.

Solution 14

The hint follows from the *triangle inequality* $|u + v| \leq |u| + |v|$ for all $u, v \in \mathbb{R}$ by setting u to $x - y$ and v to $y - z$.

We perform a proof by contradiction.

1.	Assume $L \neq M$	
2.	Let $d = L - M $	
3.	Let $\epsilon = \frac{1}{3}d$	
4.	There exists $N \in \mathbb{N}$ s.t. for all $n > N$: $ a_n - L < \epsilon$	(hyp; def limit)
5.	There exists $K \in \mathbb{N}$ s.t. for all $n > K$: $ a_n - M < \epsilon$	(hyp; def limit)
6.	Define $j = \max\{K, N\} + 1$	
7.	$ a_j - L < \epsilon$	(4, 6; us $\forall$)
8.	$ a_j - M < \epsilon$	(5, 6; us $\forall$)
9.	$ L - a_j + a_j - M < 2\epsilon$	(7, 8; prop $\mathbb{R}$)
10.	$ L - M < 2\epsilon$	(9; prop $\mathbb{R}$)
11.	$d < 2\epsilon$	(2, 10; substitution)
12.	$L = M$	(1–11, contradiction)

Exercise 15

Show: For all integers $n \geq 0$ and $0 \leq i \leq n$, $\binom{n}{i} + \binom{n}{i-1} = \binom{n+1}{i}$ holds.

Hint: $\binom{n}{i} = 0$ if $i < 0$.

Solution 15

Fix an integer n with $n \geq 0$. If $i = 0$ then $\binom{n}{i} = \binom{n}{0} = 1 = \binom{n+1}{0} = \binom{n+1}{i}$.

Now let i satisfy $0 < i \leq n$.

$$\begin{aligned}
\binom{n}{i} + \binom{n}{i-1} &= \frac{n!}{i!(n-i)!} + \frac{n!}{(i-1)!(n-i+1)!} = \frac{n!}{i!(n-i)!} + \frac{in!}{i!(n+1-i)!} \\
&= \frac{(n-i+1)n!}{i!(n-i+1)!} + \frac{in!}{i!(n+1-i)!} = \frac{n!(n-i+1+i)}{i!(n+1-i)!} = \frac{n!(n+1)}{i!(n+1-i)!} \\
&= \binom{n+1}{i}
\end{aligned}$$